

Chemical Reactions and Equations

➤ Chemical Reaction

- ✓ A chemical reaction is a process in which one or more substances are converted to one or more different substances with similar or different properties.
- ✓ These substances are either chemical elements or compounds.
- ✓ Chemical reactions involve:
 - ⊙ The breaking of existing bonds between atoms.
 - ⊙ The formation of new bonds to produce different substances.
- ✓ Some examples of chemical changes in day-to-day life:
 - ⊙ **Rusting of Iron:** Iron objects exposed to air and moisture develop a reddish-brown coating called rust.
 - ⊙ **Fermentation of Grapes:** The sugar in grapes is converted into alcohol due to microbial action.
 - ⊙ **Cooking of Food:** Heat alters the composition, taste, and texture of food, leading to irreversible changes.
 - ⊙ **Digestion of Food:** Enzymes break down complex food into simpler substances that the body can absorb.
 - ⊙ **Respiration:** Oxygen reacts with glucose to produce energy for the body.
 - ⊙ **Burning of Wood or Paper:** Wood or paper burns to produce ash, smoke, and gases.
 - ⊙ **Formation of Curd from Milk:** Lactobacillus bacteria turn milk into curd by converting lactose into lactic acid.
- ✓ A chemical reaction can be recognized through one or more of the following changes:

S. No.	Type of Change	Example of Chemical Change
1	Change in State	Cooking of an egg (proteins denature and form a new substance)
		Baking of dough (dough expands and solidifies due to chemical reactions)
		Burning of candle wax (wax vapor burns and forms new substances like CO ₂ and water)
2	Change in Color	Rusting of iron (iron turns reddish-brown due to reaction with oxygen and moisture)
		Apple turning brown when exposed to air (oxidation reaction)
		Silver articles turning black due to sulfur in the air (formation of silver sulfide)
3	Evolution of Gas	Reaction of vinegar with baking soda (carbon dioxide gas released)
		Zinc reacting with hydrochloric acid (hydrogen gas released)
		Decomposition of hydrogen peroxide (oxygen gas released)
4	Change in Temperature	Burning of magnesium ribbon (heat released in the reaction)
		Slaking of quicklime when water is added (heat released)
		Decomposition of ammonium chloride (absorbs heat, causing cooling)
5	Formation of Precipitate	Mixing of silver nitrate with sodium chloride (white precipitate of silver chloride forms)
		Reaction between lead nitrate and potassium iodide (yellow precipitate of lead iodide forms)

✓ Chemical reaction between Magnesium with Oxygen

⇒ Materials Required:

- (i) Magnesium ribbon (3-4 cm long)
- (ii) Sandpaper
- (iii) Tongs
- (iv) Spirit lamp or Bunsen burner
- (v) Watch glass
- (vi) Safety goggles

⇒ **Procedure:****1. Preparation of Magnesium Ribbon:**

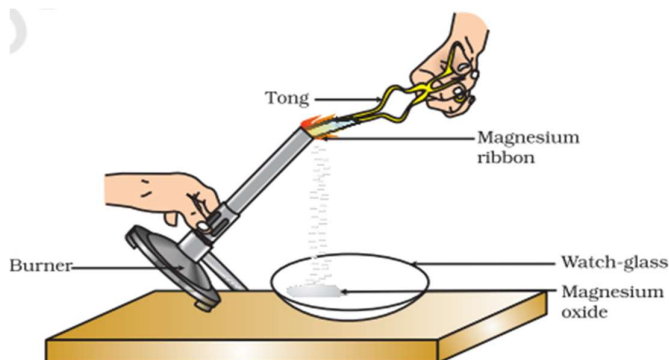
- Take a 3-4 cm long strip of magnesium ribbon.
- Rub it with sandpaper to remove any oxide layer or impurities on its surface.

2. Burning of Magnesium:

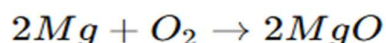
- Hold the cleaned magnesium ribbon with a pair of tongs.
- Light a spirit lamp or Bunsen burner.
- Bring the magnesium ribbon close to the flame until it catches fire.

3. Observations:

- The magnesium ribbon burns with a bright dazzling white flame.
- A white powdery substance collects as ash in the watch glass.

**4. Chemical Reaction:**

- The white powder formed is magnesium oxide (MgO).
- This is a result of magnesium reacting with oxygen from the air.

⇒ **Balanced Chemical Equation:**⇒ **Conclusion:**

- The burning of magnesium is a chemical change as a new substance (magnesium oxide) is formed.
- The reaction is exothermic, meaning it releases heat and light.

✓ **Reaction Between Lead Nitrate and Potassium Iodide**© **Materials Required:**

- Lead nitrate solution, $Pb(NO_3)_2$
- Potassium iodide solution, KI
- Test tube
- Dropper

⇒ **Procedure:**

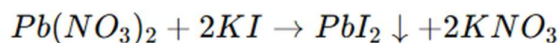
- Take about 5 mL of lead nitrate solution in a test tube.
- Using a dropper, add a few drops of potassium iodide solution to the test tube.
- Observe the changes occurring in the solution.

⇒ **Observation:**

- A bright yellow precipitate is formed immediately.

⇒ **Explanation:**

- When lead nitrate and potassium iodide solutions are mixed, a double displacement reaction (precipitation reaction) occurs.
- Lead (II) ions Pb^{2+} from lead nitrate react with iodide ions I^- from potassium iodide to form lead iodide PbI_2 , which is an insoluble yellow precipitate.

⇒ **Balanced Chemical Equation:**⇒ **Conclusion:**

- This experiment demonstrates a precipitation reaction, where two aqueous solutions react to form an insoluble solid.
- It also highlights the characteristic color of lead iodide, which is yellow.
- The formation of the yellow precipitate of lead iodide confirms the presence of lead ions in the solution.

✓ Reaction Between Zinc and Dilute Acid

⇒ Materials Required:

- Zinc granules
- Dilute Hydrochloric Acid, HCl or Dilute Sulfuric Acid, H₂SO₄
- Conical flask or test tube
- Dropper
- A burning matchstick

⇒ Procedure:

- Take a few zinc granules in a conical flask or a test tube.
- Carefully add dilute hydrochloric acid or dilute sulfuric acid using a dropper.
- Observe the reaction occurring around the zinc granules.
- Touch the test tube or flask and note any temperature change.
- Bring a burning matchstick near the mouth of the test tube.

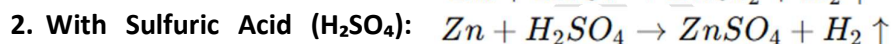
⇒ Observations:

- Effervescence (bubbling) occurs around the zinc granules.
- A colourless gas is evolved.
- When the test tube is touched, it feels warm, indicating a rise in temperature.
- When a burning matchstick is brought near the mouth of the test tube, a pop sound is heard.

⇒ Explanation:

- Zinc reacts with dilute acids like HCl or H₂SO₄ to produce hydrogen gas (H₂) and Zinc Chloride, ZnCl₂.
- The reaction releases heat energy, making it an exothermic reaction.

⇒ Chemical Equations:



⇒ Conclusion:

- This activity demonstrates a metal-acid reaction where zinc reacts with acid to produce a salt and hydrogen gas. The pop sound confirms the presence of hydrogen gas, as hydrogen burns with a small explosion.
- The release of heat makes it an exothermic reaction.

➤ Representation Chemical Reaction:

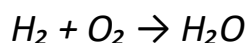
✓ Chemical Reaction can be represented by two ways:

1) Word Equation:

- ② A word equation is a simplified way to represent a chemical reaction.
- ② Reactants (substances that undergo change) are written on the left-hand side (LHS) with plus sign (+) between them.
- ② Products (new substances formed) are written on the right-hand side (RHS) with plus sign (+) between them.
- ② An arrow (→) is placed between the reactants and products, indicating the direction of the reaction.
- ② Example of Word Equation:
Magnesium + Oxygen → Magnesium Oxide [**Reactants:** Magnesium and Oxygen and **Product:** Magnesium Oxide]

2) Chemical Equation:

- ② A chemical equation is a symbolic representation of a chemical reaction using chemical formulae instead of words.
- ② Chemical equations make reactions more concise and useful by replacing names of substances with their chemical formulae.
- ② The reaction of hydrogen with oxygen to form water can be written as:



3) Balanced Chemical Equation:

- ☐ A chemical equation is balanced if the number of atoms of each element is the same on both sides of the equation.
- ☐ If the number of atoms is the same on the left-hand side (LHS) and right-hand side (RHS) of the equation, the equation is balanced.
- ☐ If the number of atoms is not equal on both sides, the equation is unbalanced. It is called as skeletal Chemical Equation.

☐ Law of Conservation of Mass:

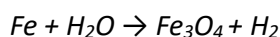
- © Mass can neither be created nor destroyed in a chemical reaction.
- © The total mass of the elements in the products must be equal to the total mass of the elements in the reactants.
- © This means that the number of atoms of each element remains the same before and after the reaction.
- ☐ An unbalanced equation does not follow the law of conservation of mass.
- ☐ A skeletal chemical equation must be balanced to satisfy the law of conservation of mass.

➤ Balancing Chemical Reaction:

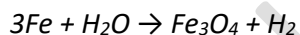
- ☐ Here are two of the best methods used to balance chemical equations:

1. Hit-and-Trial Method (Inspection Method) – Best for Simple Equations

- © This method involves adjusting the coefficients of reactants and products to balance the number of atoms on both sides.
- © Example: Formation of Fe_3O_4 (Reaction of Iron with Water)

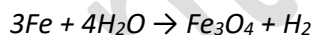
• Unbalanced Equation:**• Step-by-Step Balancing Using Hit-and-Trial**

Step 1: Balance Iron (Fe): Fe_3O_4 contains 3 Fe atoms, so we place 3 Fe on the left side:



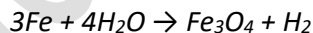
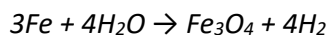
Step 2: Balance Oxygen (O): Fe_3O_4 has 4 O atoms on the right side.

- Water (H_2O) has 1 O atom per molecule.
- To balance oxygen, we need 4 H_2O molecules:

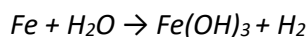


Step 3: Balance Hydrogen (H)

- $4\text{H}_2\text{O}$ molecules contribute 8 H atoms on the left.
- H_2 molecules contain 2 H atoms per molecule.
- To balance, we place 4 H_2 molecules on the right:

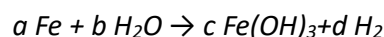
**• Final Balanced Equation:****2. Algebraic Method – Best method for Complex Equations**

- © This method assigns variables to coefficients and solves equations algebraically.
- © Example: Balancing the Formation of Ferric Hydroxide $\text{Fe}(\text{OH})_3$ from Iron and Water Using the Algebraic Method.

• Unbalanced Equation:**• Step-by-Step Balancing Using the Algebraic Method**

Step 1: Assign Coefficients

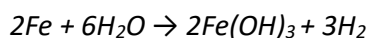
Let the balanced equation be:



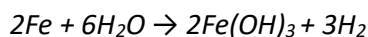
Step 2: Write separate Equations for the atoms of each Element

- **Balancing Iron (Fe):**
 - 1 Fe on the left in Fe.
 - 1 Fe on the right in $\text{Fe}(\text{OH})_3$.
 - So, $a = c$. Let $a=1$, then $c=1$
- **Balancing Oxygen (O):**
 - Each H_2O molecule contains 1 oxygen atom on the left.
 - Each $\text{Fe}(\text{OH})_3$ contains 3 oxygen atoms on the right.
 - To balance oxygen, set $b = 3$.
- **Balancing Hydrogen (H):**
 - Each H_2O has 2 hydrogen atoms, so 3 H_2O molecules contribute 6 H atoms on the left.
 - Each $\text{Fe}(\text{OH})_3$ has 3 hydrogen atoms, so 1 $\text{Fe}(\text{OH})_3$ contributes 3 H atoms on the right.
 - The remaining 3 H atoms must form H_2 molecules on the right. Since H_2 molecules have 2 H atoms per molecule, we set $d=3/2$, which is not a whole number.
 - To eliminate the fraction, multiply the entire equation by 2.

Step 3: Multiply to Remove Fractions



• Final Balanced Equation:



🔍 Writing Symbols for Physical States in a Chemical Equation:

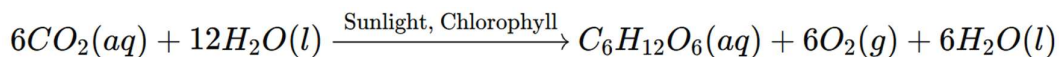
- © A balanced chemical equation usually does not indicate the physical states of reactants and products.
- © Adding physical state symbols makes the equation more informative.
- © Physical states are generally not required unless they are necessary for clarity.
- © Reaction Directions:
 - $\rightarrow$: Represents a reaction proceeding in one direction, from reactants to products.
 - $\rightleftharpoons$: Represents a reversible reaction, meaning the reaction can proceed in both directions (forward and backward).
- © Symbols for Different Physical States
 - **(s)** ---- Solid
 - **(l)** ---- Liquid
 - **(g)** or **↑**---- Gas
 - **(aq)** ---- Aqueous (dissolved in water)
 - **(ppt)** or **↓**---- Precipitate
 - Conditions like temperature, pressure, and catalysts are sometimes written above or below the reaction arrow.
 - Some reactions also mention catalysts like sunlight and chlorophyll, they are also written above or below reaction arrow.
- © Example of a Balanced Equation with Physical States
 - **Without physical states:**

$$3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$$
 - **With physical states:**

$$3\text{Fe}(s) + 4\text{H}_2\text{O}(g) \rightarrow \text{Fe}_3\text{O}_4(s) + 4\text{H}_2(g)$$

Here, (g) is used for H_2O to show that steam is used in the reaction.
 - **With necessary Conditions:**

$$\text{CO}(g) + 2\text{H}_2(g) \xrightarrow{340\text{atm}} \text{CH}_3\text{OH}(l)$$
 - **With Sunlight and Chlorophyll as catalyst:**



➤ Classification of Chemical Reaction:

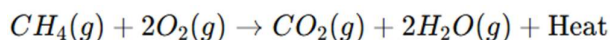
✓ Chemical reactions can be classified into different types based on various criteria.

✓ Classifications based on Energy Changes

✚ Exothermic Reaction

- © An exothermic reaction is a chemical reaction that releases heat energy into the surroundings.
- © These reactions usually involve combustion or oxidation (addition of Oxygen).
- © Exothermic Reaction can be represented by Writing “+ Heat” or “+ Energy” on the product side
- © Example:

(i) Burning of Methane(CH_4) in presence of Oxygen releases heat energy:



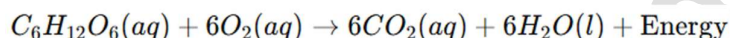
This reaction is also called as Combustion reaction.

(ii) **Decomposition of vegetable matter into compost** is also an **exothermic reaction**, as it releases heat.

(iii) Respiration as an Exothermic Process:

We get energy from food through **respiration**.

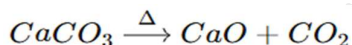
Carbohydrates (e.g., glucose) react with oxygen in our cells to produce energy.



✚ Endothermic Reaction

- © An endothermic reaction is a chemical reaction that absorbs heat energy from the surroundings.
- © These reactions require an input of heat to proceed.
- © Endothermic reaction can be represented by using the symbol “ Δ ” (delta) with an arrow
- © Example:

(i) Calcium carbonate decomposes into calcium oxide and carbon dioxide when heated.

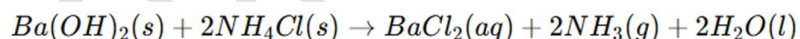


(ii) Reaction Between Barium Hydroxide and Ammonium Chloride

• Procedure:

1. Take about 2 g of barium hydroxide [$\text{Ba}(\text{OH})_2$] in a test tube.
2. Add 1 g of ammonium chloride (NH_4Cl) to the test tube.
3. Mix the two substances using a glass rod.

• Reaction:

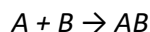


- Observation: The bottom of the test tube feels cold to the touch which indicates that it is endothermic reaction.

✓ Classification based on the Nature of Reactants and Products:

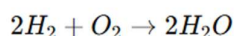
✚ Combination Reaction (Synthesis Reaction)

- © A combination reaction is a chemical reaction in which two or more reactants combine to form a single product.
- © In this type of reaction, simpler substances react to form a more complex compound.
- © These reactions are usually exothermic because they release energy. But all Combination reactions are not exothermic.
- © General Equation:



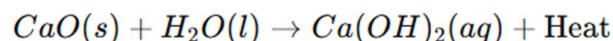
© Examples:

(i) Hydrogen and oxygen combine to form water.

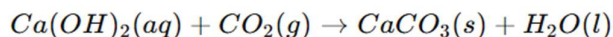


(ii) Calcium oxide (quick lime) reacts strongly with water to form slaked lime (calcium hydroxide).

- This reaction releases a heat and the product is warm.

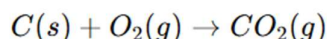


- Slaked lime when applied on walls, calcium hydroxide reacts slowly with carbon dioxide from the air.
- This forms a thin layer of calcium carbonate on the surface.



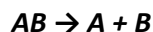
- Calcium carbonate (CaCO_3) forms within 2-3 days after whitewashing.
- It hardens and gives the walls a shiny finish. Marble is chemically the same as calcium carbonate (CaCO_3), the compound formed during whitewashing.

(iii) Burning of Carbon in presence of oxygen to produce Carbon di oxide gas:



Decomposition Reaction

- © A decomposition reaction is a chemical reaction in which a single compound breaks down into two or more simpler substances.
- © This type of reaction is the reverse of a combination reaction.
- © It often requires an external source of energy such as heat, electricity, or light.
- © Most decomposition reactions are endothermic, but all decomposition reactions are not endothermic.
- © General Equation:

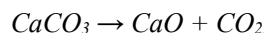


Types of Decomposition Reactions:

(i) Thermal Decomposition (heat is required)

- Thermal decomposition is a chemical reaction that occurs when a substance breaks down due to heat.
- It is also known as thermolysis.
- Thermal decomposition occurs when heat is used to break down a compound.
- Examples:

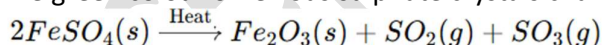
a) Thermal Decomposition of calcium carbonate:



- Calcium carbonate decomposes into calcium oxide and carbon dioxide when heated.
- Calcium oxide (quick lime) is widely used in cement manufacturing.

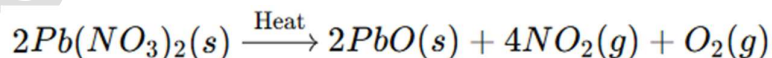
b) Thermal Decomposition of Ferrous Sulphate

- The green colour of ferrous sulphate crystals changes when heated.



- A characteristic odour of burning sulphur is observed.
- Ferrous sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) loses water upon heating, changing colour.
- It decomposes into ferric oxide (Fe_2O_3) (solid), sulphur dioxide (SO_2) (gas), and sulphur trioxide (SO_3) (gas).

c) Thermal Decomposition of Lead Nitrate



- Here, Lead Oxide and Brown fumes are emitted due to the release of nitrogen dioxide (NO_2). **Lead oxide (PbO) forms a yellow solid residue** (which is reddish-brown when hot and yellow when cold).

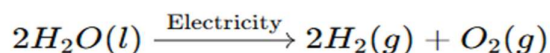
(ii) Electrolytic Decomposition (electricity is required):

- Electrolytic decomposition, also known as electrolysis, is a chemical process where electricity is used to break down a compound into simpler substances (elements or simpler compounds).
- It involves passing an electric current through a molten or dissolved ionic compound (electrolyte).
- When an electric current is passed through the electrolyte, ions (charged particles) move towards the oppositely charged electrodes (anode and cathode).

- Examples:

a) Electrolysis of Water

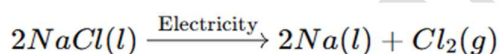
- It is breaking down of water into Oxygen and Hydrogen by electricity.
- Procedure:
 1. Set up a plastic mug with two carbon electrodes connected to a 6V battery.
 2. Fill it with water and add a few drops of dilute sulfuric acid. Pure water is a poor conductor of electricity because it has very few ions. Adding a small amount of acid (usually dilute sulfuric acid) increases the number of ions (H^+ and SO_4^{2-}) in the water. These ions help carry the electric current through the solution, making electrolysis possible.
 3. Place **two inverted test tubes** over the electrodes.
 4. Turn on the current and observe **bubble formation** at both electrodes.
- Observations:
 1. The gases collected in both test tubes differ in volume because two moles of Hydrogens produced where only one mole of Oxygen is evolved as product.
 2. When tested with a burning candle O_2 supports combustion while H_2 burns with a pop sound (H_2).
 3. Hydrogen gas (H_2) – collected at the cathode (negative electrode).
 4. Oxygen gas (O_2) – collected at the anode (positive electrode).
- Reaction:



- Conclusion: Water undergoes electrolysis to form hydrogen (H_2) and oxygen (O_2) gases.

b) Electrolysis of Molten Sodium Chloride

- When molten NaCl (not in solution, but in liquid form after melting) is electrolyzed, sodium and chlorine forms.



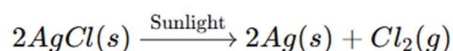
- At Cathode (–): Sodium ions (Na^+) gain electrons $\rightarrow$ Sodium metal (Na)
- At Anode (+): Chloride ions (Cl^-) lose electrons $\rightarrow$ Chlorine gas (Cl_2)

(iii) Photolytic Decomposition (light energy is required):

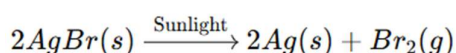
- Photolytic decomposition, also known as photolysis or photochemical decomposition, is a chemical reaction where a compound breaks down into simpler substances when exposed to light (photons).
- The energy from photons excites the molecules, causing them to break their chemical bonds and form new substances.
- Examples:

a) Photo-decomposition of Silver Chloride/Bromide:

- Silver chloride decomposes into silver and chlorine gas in the presence of sunlight.
- Procedure:
 1. Take 2 g silver chloride ($AgCl$) in a china dish.
 2. Observe its initial white color.
 3. Place the dish in sunlight for some time.
- Observation:
 1. Silver chloride turns grey due to decomposition.
- Reaction:



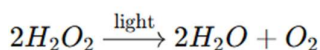
- Silver Bromide decomposes into silver and Bromine gas in the presence of sunlight.



- These reactions are used in black-and-white photography.

b) Photo-decomposition of Hydrogen Peroxide(H_2O_2):

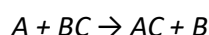
- Hydrogen peroxide is sensitive to light and can break down when exposed to sunlight or UV rays.



- Hydrogen peroxide is stored in dark-colored bottles to prevent its decomposition triggered by exposure to light.

Displacement Reaction

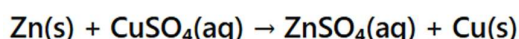
- A displacement reaction is a chemical reaction in which a more reactive element replaces a less reactive element from its compound.
- The reactivity of elements follows the reactivity series (for metals).
- A more reactive element displaces a less reactive one from its salt solution.
- General Equation:



- Example:

(i) Zinc displaces copper from copper sulphate solution.

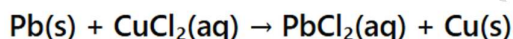
- Zinc is more reactive than copper, so it displaces copper



from copper sulphate.

(ii) Reaction between Lead and Copper Chloride:

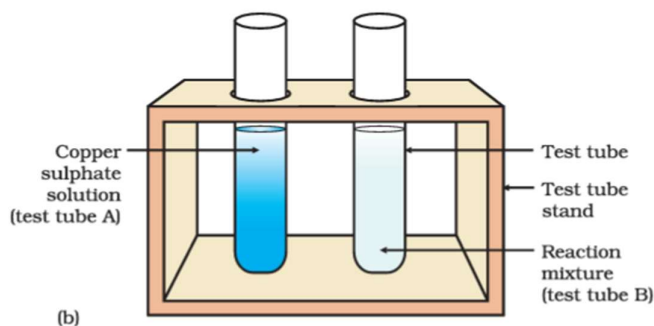
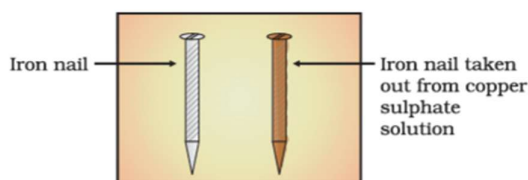
- Lead is more reactive than copper, so it displaces copper from copper chloride

**(iii) Iron nails dipped in copper sulphate solution****Materials Required:**

- Three iron nails
- Sandpaper
- Two test tubes (labelled A and B)
- Thread
- Copper sulphate solution (10 mL in each test tube)

Procedure:

- Take three iron nails and clean them thoroughly by rubbing with sandpaper to clear the layer of rust.
- Pour about 10 mL of copper sulphate solution into each of the two test tubes labelled A and B.
- Tie two iron nails together using a thread.
- Carefully immerse these nails in the copper sulphate solution in test tube B.
- Keep the third iron nail aside as a control for comparison.
- Leave the setup undisturbed for about 20 minutes.

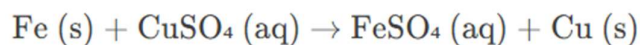


Metal	Atomic No	Symbol	Reactivity
Potassium	19	K	
Sodium	11	Na	
Lithium	3	Li	
Barium	56	Ba	
Calcium	20	Ca	
Magnesium	12	Mg	
Aluminium	13	Al	
Manganese	25	Mn	
Zinc	30	Zn	
Chromium	24	Cr	
Iron	26	Fe	
Cobalt	27	Co	
Nickel	28	Ni	
Tin	50	Sn	
Lead	82	Pb	
Hydrogen	1	H	
Copper	29	Cu	
Mercury	80	Hg	
Silver	47	Ag	
Gold	19	Au	
Platinum	78	Pt	

- **Observation After 20 Minutes:**

- The iron nails dipped in the copper sulphate solution appear brownish.
- The blue colour of copper sulphate solution in test tube B becomes fainter compared to test tube A.

- **Chemical Reaction:**



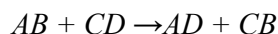
(Iron + Copper sulphate(blue) → Iron sulphate (pale green) + Copper)

- **Explanation:**

- In this reaction, iron displaces copper from copper sulphate because iron is more reactive than copper.
- The brown deposit on the iron nails is copper metal, and the fading blue colour is due to the formation of iron sulphate, which is pale green in colour.

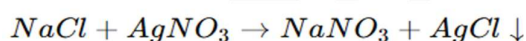
🌈 Double Displacement Reaction

- © A double displacement reaction is a chemical reaction in which two compounds exchange their ions to form two new compounds.
- © This reaction generally occurs in aqueous solutions.
- © Double displacement reaction results in an **exchange of ions** between the reactants.
- © It often results in the formation of a precipitate, a gas, or water.
- © A double displacement reaction, or metathesis reaction, typically does not occur when all the products are highly soluble or highly ionized in the solvent.
- © General Equation:

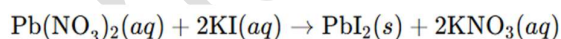


- © Example:

(i) **Sodium chloride reacts with silver nitrate to form sodium nitrate and a white precipitate of silver chloride.**



(ii) **Lead Nitrate reacts with Potassium Iodide to form Potassium Nitrate and Yellow precipitate of Lead Iodide**



(iii) **Reaction between Sodium sulphate and Barium Chloride Solution(With Lab activity)**

- **Materials Required:**

- Sodium sulphate solution (about 3 mL)
- Barium chloride solution (about 3 mL)
- Two test tubes

- **Procedure:**

- Take about 3 mL of sodium sulphate solution in one test tube.
- Take about 3 mL of barium chloride solution in another test tube.
- Mix both solutions together.

- **Observation:** A white insoluble substance (Precipitate) is formed after mixing.

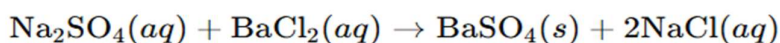
- **Explanation:**

- The precipitate formed is barium sulphate (BaSO_4).

Insoluble salts which can make Precipate in their Solution		
Salt Name	Chemical Formula	Color of Precipitate
Silver chloride	AgCl	White
Silver bromide	AgBr	Pale yellow (<i>light yellow</i>)
Silver iodide	AgI	Yellow
Barium sulphate	BaSO_4	White
Lead(II) sulphate	PbSO_4	White
Lead(II) iodide	PbI_2	Bright yellow
Calcium carbonate	CaCO_3	White
Barium carbonate	BaCO_3	White
Lead(II) carbonate	PbCO_3	White
Copper(II) hydroxide	Cu(OH)_2	Pale blue (<i>light blue</i>)
Iron(II) hydroxide	Fe(OH)_2	Green
Iron(III) hydroxide	Fe(OH)_3	Reddish brown
Zinc hydroxide	Zn(OH)_2	White
Aluminium hydroxide	Al(OH)_3	White gelatinous (<i>jelly-like</i>)
Magnesium hydroxide	Mg(OH)_2	White
Copper(II) carbonate	CuCO_3	Green
Iron(II) carbonate	FeCO_3	Pale green (<i>light green</i>)
Iron(III) carbonate	$\text{Fe}_2(\text{CO}_3)_3$	Brown

- The other product, sodium chloride (NaCl), remains dissolved in the solution.
- The reaction can be classified as a precipitation reaction because it results in the formation of an insoluble product.

• **Balanced Chemical Equation:**



(Sodium sulphate + Barium chloride → Barium sulphate + Sodium chloride)

• **Cause of the Reaction:**

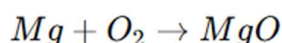
- The white precipitate of BaSO_4 is formed due to the reaction between SO_4^{2-} (sulphate ions) and Ba^{2+} (barium ions).
- The sodium (Na^+) and chloride (Cl^-) ions remain in the solution.

✓ **Classification based on Electron Transfer: Redox Reactions**

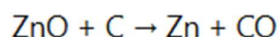
✚ **Oxidation Reaction**

- © Oxidation is a chemical reaction in which a substance gains oxygen or loses Hydrogen or electrons.
- © Oxidation reactions occur in combustion and corrosion.
- © The substance which oxidises the other substance is called Oxidising Agent.
- © It is part of a redox reaction.
- © Example:

(i) **Magnesium gains oxygen to form magnesium oxide.**



(ii) **Carbon is oxidized to Carbon monoxide when reacted with Zinc Oxide.**



(iii) **Oxidation of Copper-to-Copper Oxide**

• **Materials Required:**

- Copper powder (about 1 g)
- China dish
- Heat source

• **Procedure:**

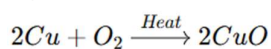
- Place the copper powder in the china dish.
- Heat it strongly in the presence of air (oxygen).

• **Observation:**

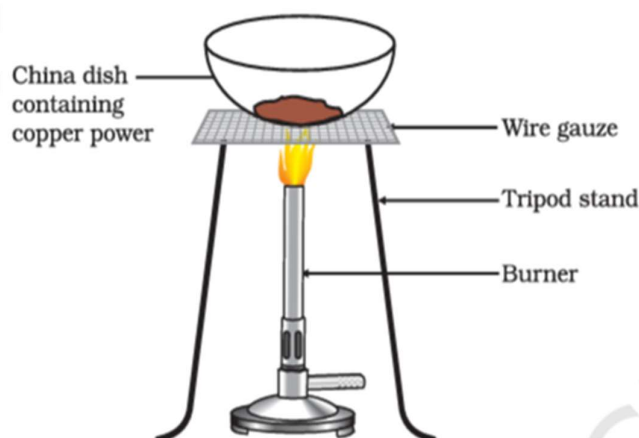
- The surface of the copper powder turns black.
- The black coating is copper(II) oxide (CuO).

• **Explanation:**

- Copper reacts with oxygen during heating:



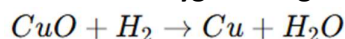
- Oxygen is added to copper → oxidation.



✚ **Reduction Reaction**

- © Reduction is a chemical reaction in which a substance loses oxygen or gains Hydrogen or electrons.
- © The substance which reduces the other substance is called **Reducing Agent**.
- © Example:

(i) **Copper (II) oxide loses oxygen and gets reduced to copper.**

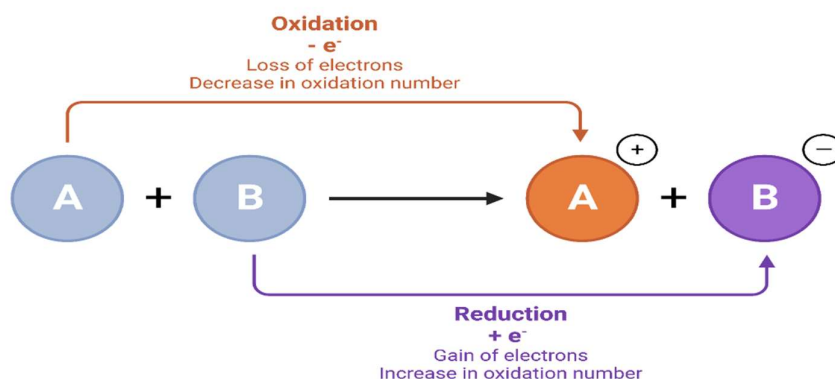


(ii) **Manganese dioxide when reacted with Hydrochloric acid gets reduced to Manganese Chloride**



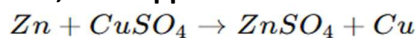
✚ **Redox Reaction**

- © A redox reaction is a chemical reaction in which oxidation and reduction occur simultaneously.
- © Redox reactions involve the transfer of electrons between chemical species.
- © Many reactions, like acid-base neutralization and precipitation reactions, are not redox reactions because they don't involve electron transfer.



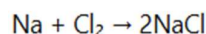
© Example:

(i) Zinc is oxidized, and copper is reduced.



- Here Zn is oxidised and Cu is reduced.

(ii) Reaction between Sodium and Chlorine to form Sodium Chloride

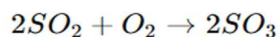


- Here Na is Oxidised as Na loses electron where Cl is reduced as it gains electron.

✓ Based on the Phase of Reactants

✚ Homogeneous Reaction

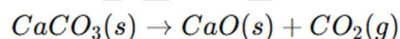
- © A homogeneous reaction is a chemical reaction in which the reactants and products are in the same phase (solid, liquid, or gas).
- © Example: Sulphur dioxide turns to sulphur trioxide on oxidation.



All reactants and products of this reaction are gases.

✚ Heterogeneous Reaction

- © A heterogeneous reaction is a chemical reaction in which the reactants are in different phases.
- © Example: Calcium carbonate (solid) decomposes into calcium oxide (solid) and carbon dioxide (gas).



➤ Oxidation Reaction in Everyday Life

- ✓ Effects of Oxidation can also be seen in daily life.
- ✓ Corrosion and Rancidity are two important effects.

✓ Corrosion:

- © Corrosion can be defined as the degradation of a metal due to chemical reaction with the substances present in environment is called corrosion.
- © When a metal is attacked by substances in its surroundings (e.g., moisture, acids, air), it is said to corrode.
- © Degradation of metal implies deterioration of physical properties of the material.
- © Iron railings, car bodies, bridges, ships, and many other metal objects, especially those made of iron, are affected by Corrosion.

© Rusting:

- Rusting is the process where iron (or iron-containing alloys like steel) corrodes, forming a reddish-brown flaky substance called rust due to a reaction with oxygen and water.
- Rusting requires the presence of both oxygen and moisture (water or water vapor).
- Rust is hydrated iron(III) oxide with formulae $Fe_2O_3 \cdot xH_2O$

- Iron articles are shiny when new.
- Over time, they become coated with a reddish-brown powder.
- This process is commonly known as rusting of iron.

© **Corrosion of Copper:**

- Copper, when exposed to the environment, reacts with oxygen and moisture, leading to the formation of copper oxide, a dull brown or black layer on the surface.
- Reaction with carbon dioxide, can result in the formation of copper carbonate which appears as a green or greenish-blue film. It is a mixture of copper carbonate CuCO_3 and copper hydroxide $\text{Cu}(\text{OH})_2$.
- Copper tarnishes when exposed to some sulphur compounds, with which it reacts to form various copper sulphides.

© **Corrosion of Silver:**

- Silver corrosion or tarnishing, occurs when silver reacts with sulphur-containing gases in the air.
- This reaction forms a black or brown layer of silver sulphide (Ag_2S) on the surface of the silver.

© **Impact of Corrosion:**

- Corrosion of iron is a serious problem.
- A huge amount of money is spent every year to replace or repair damaged iron objects.

© **Methods of Prevention from Corrosion**

1. Painting

- Coating the metal surface with paint prevents contact with air and moisture, which are necessary for corrosion.
- Commonly used on iron railings, fences, and vehicles.

2. Oiling or Greasing

- Applying oil or grease forms a protective layer on metal surfaces.
- Especially useful for moving parts of machines like bike chains or tools.

3. Galvanization

- Coating iron or steel with a layer of zinc.
- Zinc prevents rusting by acting as a sacrificial metal, corroding in place of iron.
- Common in water pipes, buckets, and roofing materials.

4. Electroplating

- Covering metal objects with a thin layer of a less reactive metal (e.g., chromium, nickel, or silver).
- Improves appearance and resists corrosion.
- Used in jewellery, cutlery, and car parts.

5. Alloying

- Mixing metals to form alloys like stainless steel (iron + chromium + nickel).
- Alloys are more resistant to corrosion than pure metals.

✓ **Rancidity**

- © Rancidity is a chemical process that occurs when fats and oils in food oxidize, which can cause food to smell and taste unpleasant. Rancid food can be unsafe to eat.
- © This leads to a change in taste and smell of the food.
- © It occurs when food containing fats or oils is exposed to air for a long time.
- © **Chips, biscuits, ghee, or fried items** left open for too long start to smell and taste bad.
- © **Prevention of Rancidity:** Various methods used for prevention are mentioned.

1. Storing in Airtight Containers

- It keeps oxygen and moisture from the air away from food item.
- It Slows down or prevents oxidation of oils and fats.
- Example: Ghee or oil stored in tightly sealed jars.

2. Refrigeration (Cold Storage)

- It lowers the temperature, which reduces the rate of oxidation.
- It keeps the food stay fresh for a longer time.
- Example: Butter or cream kept in the refrigerator.

3. Using Anti-oxidants

- Anti-Oxidants are Chemical substances that prevent oxidation.
- BHA (Butylated hydroxyanisole) and BHT (Butylated hydroxytoluene) are commonly used Anti-Oxidants
- They are added to processed and packaged food to extend shelf life.

4. Packing with Inert Gas (e.g., Nitrogen)

- Nitrogen gas is filled in the food packet to push out oxygen.
- Example: Packets of potato chips are filled with nitrogen to keep them fresh and crisp.

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